

Week 8: Systemd & Service Management (Part 2)

Date: April 27, 2026

Study Time: 5-8 hours total

GOAL

Create and manage custom systemd services

STUDY TASKS (1-2 hours)

Reading

☐ "How Linux Works" - Chapter 6 (systemd unit files)

Video Learning

☐ LearnLinuxTV - "Create Systemd Service"

☐ Professor Messer - "Systemd Unit Files"

LAB WORK (3-4 hours)

Create Custom Service

```
bash

# Create service file
sudo nano /etc/systemd/system/myservice.service
```

Basic Unit File Structure

ini

```
[Unit]
Description=My Custom Service
After=network.target

[Service]
Type=simple
User=username
WorkingDirectory=/home/username
ExecStart=/path/to/script.sh
Restart=on-failure
RestartSec=5s

[Install]
WantedBy=multi-user.target
```

Example: Monitoring Script Service

```
bash

# 1. Create script
nano ~/monitoring-script.sh

#!/bin/bash
while true; do
    echo "$(date): System check" >> /var/log/my-monitor.log
    sleep 60
done

# 2. Make executable
chmod +x ~/monitoring-script.sh

# 3. Create service
sudo nano /etc/systemd/system/monitor.service
```

ini

[Unit]

Description=System Monitoring Script

After=network.target

[Service]

Type=simple

User=your_username

ExecStart=/home/your_username/monitoring-script.sh

Restart=always

[Install]

WantedBy=multi-user.target

Manage Custom Service

bash

Reload systemd to recognize new service

sudo systemctl daemon-reload

Start service

sudo systemctl start monitor.service

Check status

systemctl status monitor.service

View logs

journalctl -u monitor.service -f

Enable at boot

sudo systemctl enable monitor.service

Enable and start

sudo systemctl enable --now monitor.service

Lab Tasks

- ☐ Create custom service for monitoring
- ☐ Create service for Gitea backup script
- ☐ Set up auto-restart on failure
- ☐ Practice viewing logs with journalctl
- ☐ Test service survives reboot

FLASHCARD COMMANDS

1. `systemctl daemon-reload`
 2. `journalctl -u`
 3. `systemctl cat SERVICE`
 4. `systemctl edit SERVICE`
 5. `systemctl enable --now`
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EXAM OBJECTIVES

[https://partners.comptia.org/docs/default-source/resources/comptia-linux-xk0-005-exam-objectives-\(3-0\)](https://partners.comptia.org/docs/default-source/resources/comptia-linux-xk0-005-exam-objectives-(3-0))

NOTES

Custom services created:

 **END OF MONTH 2 - Foundations Complete!**

Completed: ☐

Previous Week: Systemd Basics

Next Week: Networking Fundamentals